

PySpark for Data Engineering — Zero to Advanced Roadmap

Your background: Python + SQL, new to Spark **Practice environment:** Local PySpark (moving to Databricks/cloud for Module 7+)

Module 1 — Spark Fundamentals

What Spark is and why it exists (vs. pandas, vs. Hadoop MapReduce)

Cluster architecture: Driver, Executors, Cluster Manager

SparkSession, SparkContext

Lazy evaluation, transformations vs. actions

DataFrames vs. RDDs (why DataFrames win for data engineering)

Module 2 — DataFrame Core Operations

Creating DataFrames (from files, from Python data, with schemas)

select, filter, withColumn, drop, rename

Aggregations: groupBy, agg

Sorting, distinct, limit

Joins (inner, left, right, outer, semi, anti) and join strategies

Module 3 — Reading & Writing Data

File formats: CSV, JSON, Parquet, ORC, Avro

Schema inference vs. explicit schemas (StructType)

Partitioned writes, write modes

Handling nested/semi-structured data (structs, arrays, maps)

Module 4 — Spark SQL

Temp views, spark.sql()

Window functions (rank, row_number, lead/lag, running totals)

UDFs vs. built-in functions (and why to avoid UDFs when possible)

Pandas UDFs (vectorized UDFs)

Module 5 — Performance & Internals

Catalyst optimizer & physical plans (explain())

Partitions, repartition vs. coalesce

Shuffles — what causes them, how to reduce them

Broadcast joins vs. shuffle joins

Caching/persistence strategies

Skew handling (salting techniques)

Module 6 — Data Engineering Patterns

Building idempotent, incremental ETL pipelines

Slowly Changing Dimensions (SCD Type 1 & 2) in Spark

Data quality checks & validation

Delta Lake: ACID transactions, time travel, MERGE INTO, schema evolution

Medallion architecture (Bronze/Silver/Gold)

Module 7 — Structured Streaming

Micro-batch vs. continuous processing

Reading streams (files, Kafka)

Watermarking, windowed aggregations

Streaming joins, checkpointing, fault tolerance (*This is where we move to Databricks Community Edition or a cloud sandbox*)

Module 8 — Production & Orchestration

Cluster sizing & configuration tuning (executor memory/cores)

Job orchestration (Airflow / Databricks Workflows)

Monitoring via Spark UI

CI/CD for Spark jobs, testing with pytest + chispa/pyspark-test

Cost optimization on cloud platforms

Module 9 — Advanced / Interview-Ready Topics

Custom partitioners, custom data sources

Adaptive Query Execution (AQE)

Memory management deep dive (execution vs. storage memory)

Common data engineering interview problems solved in PySpark

How we'll work: Each module = short concept explanation + runnable code we execute together + a small exercise for you to try. We move at your pace — say "next" to continue or ask questions anytime.